Finite-Time Tracking of Arbitrary Swarm Formations with Safety and Connectivity Certificates

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***Abstract*— Certifying finite arrival times and convergence rates is essential for coordinating multi-agent systems in time-critical missions. However, convergence to a desired formation does not, by itself, guarantee collision avoidance or communication connectivity during transients. Existing swarm models can achieve finite-time tracking of collective variables, but may not provide independent settling guarantees for agents at arbitrarily prescribed stations. This paper proposes a first-order control framework for tracking arbitrary, time-varying formation stations relative to a moving reference. A component-wise fractional-power feedback ensures exact finite-time station tracking with an explicit settling-time bound independent of the interaction graph. To address transient safety, we derive a graph-dependent exponential bound on relative disagreement that remains valid for time-varying and switching graphs, including disconnected modes. Combining this bound with the geometry of the prescribed stations yields sufficient conditions for pairwise collision avoidance and preservation of a connected spanning communication backbone. These conditions further provide a coupling-gain design rule based on the available geometric clearance throughout the maneuver. Numerical simulations illustrate finite-time tracking, transient formation deformation, and the effectiveness and conservatism of the proposed safety and connectivity certificates.**

***Index Terms*— Multi-agent systems, formation control, finite-time stability, collision avoidance, network connectivity.**

I. Introduction

Multi-agent systems enable coordinated tasks in which individual robots must move collectively while maintaining a prescribed spatial arrangement [1]. In many applications, the desired formation is asymmetric or time-varying, and each agent must reach its assigned station relative to a moving reference. For time-critical missions, asymptotic convergence may be insufficient: all agents must reach their stations within a finite time, without colliding or losing essential communication links during the transient. These requirements are coupled, since tracking errors can cause collisions or push communicating agents beyond sensing range even when the desired formation is itself feasible.

Finite-time stability provides exact convergence within a finite settling time [3]. However, finite-time convergence alone does not certify that the transient remains physically feasible. Existing formation-control approaches address related objectives through different mechanisms. For example, prescribed-performance distance-based formation control shapes formation errors [5], while barrier-function methods have been used to jointly enforce collision avoidance and connectivity maintenance for constrained second-order agents [**?**]. The problem considered here is complementary: to obtain an explicit, graph-independent deadline for tracking individually prescribed stations and use a separate analytical bound to certify collision avoidance and persistence of essential communication links.

A particularly relevant starting point is the kinematic swarm model of Fedele and D'Alfonso [4], which achieves finite-time centroid tracking and finite-time aggregation around the centroid trajectory. Its interaction dynamics determine the resulting aggregation, however, rather than allowing each agent's station to be prescribed independently. Moreover, the centroid-level convergence result does not itself provide a finite-time settling guarantee for every agent's error relative to an arbitrary assigned station. These gaps motivate separating station tracking from the analysis of relative agent motion.

This paper develops a first-order controller for tracking arbitrary prescribed station trajectories. A component-wise fractional-power feedback yields an explicit finite-time settling bound independent of the interaction graph. A second, graph-dependent estimate bounds relative disagreement for undirected time-varying or switching graphs, including disconnected modes. Combining this estimate with the geometry of the desired stations gives sufficient conditions for collision avoidance and preservation of a designer-selected connected spanning communication backbone. The conditions further yield a coupling-gain inequality based on the available time-varying geometric clearance.

The main contributions are:

1) A first-order station-tracking controller with an explicit, graph-independent finite-time settling bound for arbitrary prescribed C^1 station trajectories.
2) A graph-dependent exponential disagreement bound valid for undirected time-varying or switching graphs, including disconnected modes.
3) Sufficient geometric conditions for pairwise collision avoidance and persistence of a connected spanning communication backbone throughout the transient.
4) A coupling-gain design inequality derived from the time-varying clearance margin, illustrated through numerical examples that distinguish formal certification from observed safe behavior.

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II. Background and Problem Statement

III. Model and Control Law

Consider $n \geq 2$ agents with first-order dynamics

$$\dot{x}_i = u_i, \qquad i = 1, \ldots, n, \tag{1}$$

where $x_i \in \mathbb{R}^d$ and $u_i \in \mathbb{R}^d$ denote the position and control input of agent i, respectively. To achieve finite-time tracking of arbitrary time-varying formations, we consider the distributed controller

$$u_i = \dot{q}_i - \delta_i - \Phi_\beta(\delta_i) - \sum_{j=1}^{n} w_{ij} M(\delta_i - \delta_j), \tag{2}$$

where q_i is the desired station of agent i, $\delta_i = x_i - q_i$ is its tracking error, Φ_β is a component-wise fractional-power feedback, w_{ij} are nonnegative interaction weights, and $M \in \mathbb{R}^{d\times d}$ is a symmetric positive semidefinite coupling matrix.

The desired station trajectories are defined as

$$q_i(t) = r(t) + p_i(t), \qquad \delta_i(t) = x_i(t) - q_i(t), \tag{3}$$

where $r(t) \in \mathbb{R}^d$ is a reference trajectory and $p_i(t) \in \mathbb{R}^d$ is the desired offset of agent i. Unless stated otherwise, the station trajectories q_i are continuously differentiable with locally bounded derivatives. This formulation allows the desired formation to translate and change shape over time. For $0 < \beta < 1$, the component-wise fractional-power feedback is defined by

$$[\Phi_\beta(z)]_k = \operatorname{sign}(z_k)|z_k|^\beta, \qquad k = 1, \ldots, d, \tag{4}$$

with $\Phi_\beta(0) = 0$.

The interactions among agents are represented by an undirected weighted graph with symmetric weights $w_{ij} = w_{ji} \geq 0$ and $w_{ii} = 0$. Let $W = [w_{ij}]$ denote the weighted adjacency matrix and define the weighted graph Laplacian as

$$L_w = \operatorname{diag}\left(\sum_{j=1}^{n} w_{ij}\right) - W. \tag{5}$$

For the range-dependent interaction model considered in this work, the weights are given by

$$w_{ij}(x,t) = \frac{a_{ij}(x,t)}{\|x_i - x_j\|^\nu + \varepsilon_w}, \qquad \nu \geq 1, \quad \varepsilon_w > 0, \tag{6}$$

where $a_{ij} = a_{ji} \in \{0,1\}$ indicates whether agents i and j are connected, and $a_{ii} = 0$. The positive constant ε_w prevents the weights from becoming singular when agents coincide.

Using the station-error definition in (3), we have

$$\delta_i - \delta_j = (x_i - x_j) - (q_i - q_j). \tag{7}$$

Thus, the coupling term can be computed from relative agent positions and the desired relative station assignments. Substituting the controller (2) into the agent dynamics (1) yields the closed-loop error dynamics

$$\dot{\delta} = -\delta - \Phi_\beta(\delta) - (L_w(x,t) \otimes M)\delta, \tag{8}$$

where $\delta = [\delta_1^\top, \ldots, \delta_n^\top]^\top$ and $\Phi_\beta(\delta)$ denotes the stacked component-wise fractional-power feedback.

Assumption 1. *The weights are locally Lipschitz in the agent positions and locally bounded between topology switches. The switching rule is well posed, has finitely many switches on every bounded time interval, and the agent states remain continuos throughout.*

IV. Finite-Time Station Tracking and Reference Convergence

Let $\delta_0 = \delta(t_0)$ and define the non-negative envelope

$$\mathcal{R}(t) = \left[\left(1 + \|\delta_0\|^{1-\beta}\right)e^{-(1-\beta)(t-t_0)} - 1\right]^{1/(1-\beta)}, \tag{9}$$

and finite-time arrival deadline be

$$T = t_0 + \frac{\ln(1 + \|\delta_0\|^{1-\beta})}{1-\beta}. \tag{10}$$

Theorem 1 (Finite-time assigned-station tracking)**.** *Under Assumption 1, with $M = M^\mathsf{T} \succeq 0$ and $0 < \beta < 1$, every solution of* (8) *satisfies*

$$\|\delta(t)\| \leq \mathcal{R}(t), \qquad t \geq t_0, \tag{11}$$

and $\delta(t) = 0$ for all $t \geq T$. Thus, $x_i(t) = q_i(t)$ for all agents and all $t \geq T$. The deadline is independent of the interaction graph, its connectivity, and M.

Proof. Consider $V = \frac{1}{2}\|\delta\|^2$. Symmetry of the weights and positive semi-definiteness of M give the edge-energy identity

$$\begin{aligned}\mathcal{D} &:= \delta^\mathsf{T}(L_w \otimes M)\delta \\ &= \frac{1}{2}\sum_{i,j} w_{ij}(\delta_i - \delta_j)^\mathsf{T} M(\delta_i - \delta_j) \geq 0.\end{aligned} \tag{12}$$

Since $1+\beta < 2$, norm monotonicity implies $\sum_{\ell=1}^{nd} |\delta_\ell|^{1+\beta} \geq \|\delta\|_2^{1+\beta}$. Consequently along (8),

$$\dot{V} = -\|\delta\|^2 - \sum_{i=1}^{n}\sum_{k=1}^{d} |\delta_{ik}|^{1+\beta} - \mathcal{D} \tag{13}$$

$$\leq -\|\delta\|^2 - \|\delta\|^{1+\beta} = -2V - (2V)^{(1+\beta)/2}, \tag{14}$$

Let $z = \|\delta\| = \sqrt{2V}$. Whenever $z > 0$, (14) implies $\dot{z} \leq -z - z^\beta$. With $y = z^{1-\beta}$,

$$\dot{y} \leq -(1-\beta)(y+1). \tag{15}$$

Integration yields $y(t) + 1 \leq (y(t_0) + 1)e^{-(1-\beta)(t-t_0)}$ until the tracking error reaches zero. This comparison bound establishes the bounds in (9) and (10), guaranteeing that the error reaches zero within a finite time. Since the origin is invariant under the error dynamics, the error remains zero after reaching it. Since V is non-increasing, the error remains bounded; the regularity and well-posed switching assumptions ensure the comparison holds across topology changes. □

A. Centroid and Formation Guarantees

Assume the station offsets are centered:

$$\sum_{i=1}^{n} p_i(t) = 0, \qquad \forall t \geq t_0. \tag{16}$$

Let $c(t) = \frac{1}{n}\sum_i x_i(t)$ be the swarm centroid. Since $q_i = r + p_i$, the mean desired station is r.

Corollary 1 (Reference tracking)**.** *Under the centering condition* (16)*,*

$$\|c(t) - r(t)\| \leq \frac{\mathcal{R}(t)}{\sqrt{n}}, \qquad t \geq t_0. \tag{17}$$

Consequently, $c(t) = r(t)$ *for all* $t \geq T$.

Proof. Since $q_i = r + p_i$ and $\sum_{i=1}^{n} p_i = 0$, the centroid error satisfies

$$c - r = \frac{1}{n}\sum_{i=1}^{n}(x_i - q_i) = \frac{1}{n}\sum_{i=1}^{n}\delta_i.$$

By the Cauchy–Schwarz inequality,

$$\|c - r\| \leq \frac{1}{\sqrt{n}}\left(\sum_{i=1}^{n}\|\delta_i\|^2\right)^{1/2} = \frac{\|\delta\|}{\sqrt{n}}.$$

Applying the bound $\|\delta(t)\| \leq \mathcal{R}(t)$ from Theorem 1 gives the stated inequality. Since $\mathcal{R}(t) = 0$ for all $t \geq T$, it follows that $c(t) = r(t)$ for all $t \geq T$. □

Remark 1. *The finite-time reference result is inherited from exact station tracking. If the offsets are not centered, the station theorem remains valid but the swarm centroid tracks the mean station* $r + \frac{1}{n}\sum_i p_i$*, not* r.

V. Disagreement-Based Safety and Connectivity Certificates

The complete station error contains both a common component and a disagreement component. Only disagreement changes relative positions. Define

$$\begin{aligned} &\Pi = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^{\mathsf{T}}, \qquad \eta = (\Pi \otimes I_d)\delta, \\ &\mu = \lambda_{\min}(M) \geq 0. \end{aligned} \tag{18}$$

Thus $\eta_i = \delta_i - \bar{\delta}$ and $\delta_i - \delta_j = \eta_i - \eta_j$.

Proposition 1 (Graph-dependent disagreement contraction)**.** *For the undirected weighted graph in* (6)*, every solution satisfies*

$$\|\eta(t)\| \leq \|\eta(t_0)\| \exp\left[-\int_{t_0}^{t}\left(1 + \mu\lambda_2(L_w(s))\right) ds\right] \tag{19}$$

Here $\lambda_2(L_w)$ *is the algebraic connectivity, with* $\lambda_2 = 0$ *for a disconnected graph and* $\mu = \lambda_{\min}(M)$*.The bound remains valid through admissible topology switches.*

Proof. Since $\Pi L_w = L_w$ and $\Pi\eta = \eta$, the disagreement dynamics followed from (8) are

$$\dot{\eta} = -\eta - (\Pi \otimes I_d)\Phi_\beta(\delta) - (L_w \otimes M)\eta. \tag{20}$$

For $W_\eta = \frac{1}{2}\|\eta\|^2$, use $\eta^{\mathsf{T}}(\Pi \otimes I_d) = \eta^{\mathsf{T}}$ to obtain

$$\dot{W}_\eta = -\|\eta\|^2 - \eta^{\mathsf{T}}\Phi_\beta(\delta) - \eta^{\mathsf{T}}(L_w \otimes M)\eta. \tag{21}$$

The non-linear term is non-negative because component-wise monotonicity of Φ_β implies

$$\eta^{\mathsf{T}}\Phi_\beta(\delta) = \frac{1}{2n}\sum_{i,j}(\delta_i - \delta_j)^\top\big(\Phi_\beta(\delta_i) - \Phi_\beta(\delta_j)\big) \geq 0 \tag{22}$$

Since $M \succeq 0$ and $\mu = \lambda_{\min}(M)$, the Rayleigh quotient bound gives

$$\eta^\top(L_w \otimes M)\eta \geq \mu\lambda_2(L_w)\|\eta\|^2$$

Discarding the non-positive term,

$$\dot{W}_\eta \leq -2\big[1 + \mu\lambda_2(L_w)\big]W_\eta.$$

Applying the comparison lemma and integrating yields (19). □

A. Pairwise Distance Bounds

Let $d_{ij}^q(t) = \|q_i(t) - q_j(t)\|$ and $d_{ij}^x(t) = \|x_i(t) - x_j(t)\|$.

Lemma 1 (Relative-distance envelope)**.** *For every pair* i, j*,*

$$d_{ij}^q(t) - \sqrt{2}\,\|\eta(t)\| \leq d_{ij}^x(t) \leq d_{ij}^q(t) + \sqrt{2}\,\|\eta(t)\|\,. \tag{23}$$

Proof. Since $x_i - x_j = (q_i - q_j) + (\eta_i - \eta_j)$, the triangle inequality gives

$$\|x_i - x_j\| \leq \|q_i - q_j\| + \|\eta_i - \eta_j\|,$$

while the reverse-triangle inequality yields

$$\|x_i - x_j\| \geq \|q_i - q_j\| - \|\eta_i - \eta_j\|.$$

Moreover,

$$\|\eta_i - \eta_j\| \leq \|\eta_i\| + \|\eta_j\| \leq \sqrt{2}(\|\eta_i\|^2 + \|\eta_j\|^2) \leq \sqrt{2}\,\|\eta\|\,.$$

Hence, the two inequalities above establishes the claimed bounds. □

To preserve communication links that are critical for coordination during formation maneuvers, we consider a designer-selected connected spanning backbone $\mathcal{H} = (\mathcal{V}, \mathcal{E}_\mathcal{H})$. Let $L_\mathcal{H}$ denote the unweighted Laplacian of $\mathcal{H}$. We define the station-edge length envelope and the minimum pairwise station distance as

$$\begin{aligned} \ell_\mathcal{H}(t) &= \max_{(i,j)\in\mathcal{E}_\mathcal{H}} \|q_i(t) - q_j(t)\|\,, \\ d_q(t) &= \min_{i<j} \|q_i(t) - q_j(t)\|\,. \end{aligned} \tag{24}$$

Let $\lambda_2(L_\mathcal{H}) > 0$ and let $\underline{w} > 0$ be a lower bound on every active backbone-edge weight. For the range-dependent weights in (6), one may choose $\underline{w} = (R_c^\nu + \varepsilon_w)^{-1}$. Define

$$\begin{aligned} \gamma_\mathcal{H} &= 1 + \mu\underline{w}\lambda_2(L_\mathcal{H}), \\ \rho_0 &= \sqrt{2}\,\|\eta(t_0)\|\,. \end{aligned} \tag{25}$$

Since $\eta = (\Pi \otimes I_d)\delta$ and Π is an orthogonal projection, it cannot increase the Euclidean norm; hence $\|\eta(t)\| \leq \|\delta(t)\|$.

By Theorem 1, this gives $\sqrt{2}\|\eta(t)\| \le \sqrt{2}\,\mathcal{R}(t)$. Moreover, Proposition 1 gives exponential contraction of the disagreement. While all backbone edges remain active, $L_w \succeq \underline{w} L_{\mathcal{H}}$, so the contraction rate is at least $\gamma_{\mathcal{H}}$. Consequently,

$$\rho_{\mathcal{H}}(t) = \min\left\{\sqrt{2}\,\mathcal{R}(t),\ \rho_0 e^{-\gamma_{\mathcal{H}}(t-t_0)}\right\}. \tag{26}$$

This is an a priori envelope on the maximum pairwise disagreement for every pair of agents, provided the backbone remains available.

Theorem 2 (Joint collision avoidance and connectivity preservation)**.** *If the planned stations satisfy, for all $t \ge t_0$,*

$$d_q(t) > d_{\text{safe}} + \rho_{\mathcal{H}}(t), \tag{27}$$

$$\ell_{\mathcal{H}}(t) < R_c - \rho_{\mathcal{H}}(t), \tag{28}$$

then every pair of agents remains strictly separated by more than d_{safe}, every backbone edge remains within range R_c, and the communication graph remains connected. The station-tracking deadline T in (10) *is unchanged.*

Proof. At t_0, (26) gives $\rho_{\mathcal{H}}(t_0) = \rho_0$, since $\|\eta(t_0)\| \le \|\delta(t_0)\|$. By Lemma 1 and (27), all actual pairwise distances exceed d_{safe} initially. The same lemma and (28) put all backbone links strictly inside R_c initially.

Suppose a backbone edge first reaches R_c at time $t_e > t_0$. Until t_e, all backbone edges remain active, so the lower bound $L_w \succeq \underline{w} L_{\mathcal{H}}$ and Proposition 1 imply (26). By continuity, the bound also holds at t_e. For every backbone edge, Lemma 1 then gives $\|x_i(t_e) - x_j(t_e)\| \le \ell_{\mathcal{H}}(t_e) + \rho_{\mathcal{H}}(t_e) < R_c$, a contradiction. Thus the backbone persists for all time. Finally, the lower half of Lemma 1 and (27) imply $d_{ij}^x(t) > d_{\text{safe}}$ for every pair and all times. The station-tracking theorem is unaffected by the graph argument. □

B. Coupling-Gain Design Rule

Define the available geometric margin

$$s(t) = \min\{d_q(t) - d_{\text{safe}},\ R_c - \ell_{\mathcal{H}}(t)\}.$$

The station plan is geometrically feasible for the selected threshold and backbone only if $s(t) > 0$ throughout the maneuver. Suppose $s(t) > 0$ on $[t_0, T]$ and $0 < \rho_0 < s(t_0)$. Define

$$\gamma_{\text{req}} = \sup_{t_0 < t < T} \frac{\ln\big(\rho_0 / s(t)\big)}{t - t_0}. \tag{29}$$

Corollary 2 (Sufficient coupling selection)**.** *A sufficient condition for Theorem 2 is*

$$1 + \lambda_{\min}(M)\,\underline{w}\,\lambda_2(L_{\mathcal{H}}) > \gamma_{\text{req}}. \tag{30}$$

Proof. By Theorem 2, it suffices to ensure that $\rho_{\mathcal{H}}(t) < s(t)$ throughout the maneuver. By the definition of γ_{req} in (29), the condition $\gamma_{\mathcal{H}} > \gamma_{\text{req}}$ ensures that

$$\rho_0 e^{-\gamma_{\mathcal{H}}(t-t_0)} < s(t), \qquad t \in (t_0, T).$$

Since from (26), $\rho_{\mathcal{H}}(t) \le \rho_0 e^{-\gamma_{\mathcal{H}}(t-t_0)}$, the collision and range conditions hold on this interval. The strict initial condition handles $t = t_0$. At and after T, $\mathcal{R}(t) = 0$ and hence $\rho_{\mathcal{H}}(t) = 0 < s(t)$, so the geometric conditions suffice. Thus, the sufficient conditions of Theorem 2 are satisfied. □

Remark 2. *If $\gamma_{\text{req}} < 1$, the baseline local term already satisfies the rate condition with $M = 0$. The rule is sufficient, not necessary or optimal: failure of the inequality means that this certificate cannot conclude safety, not that a collision must occur. Increasing coupling cannot repair a failed initial-margin test $\rho_0 < s(t_0)$.*

TABLE I: Benchmark simulation parameters.

Param.	Value	Description
n	10	Swarm size in $\mathbb{R}^2$
β	0.5	Fractional-power velocity exponent
d_{safe}	1.0 m	Minimum pairwise collision threshold
R_c	6.0 m	Maximum communication sensing radius
ε_w	0.05	Proximity weight regularizer
$\underline{w}$	0.1653 m^{-1}	Uniform lower bound $1/(R_c + \varepsilon_w)$
$r(t)$	$[t,\ 2\sin(0.5t)]^{\mathsf{T}}$	Reference centroid trajectory
ω	$\pi/4$ rad/s	Star template angular rotation rate
M	mI_2	Cooperative coupling matrix ($m \ge 0$)

VI. Numerical Validation

The theoretical results are evaluated through numerical simulations for $n = 10$ agents in $\mathbb{R}^2$ executing dynamic maneuvers under the benchmark parameters summarized in Table I. To facilitate reproducibility, the simulation framework and complete experiment suite are openly available at `https://github.com/yash-purswani/finite-time-swarm-control`.

Three scenarios are investigated: (i) finite-time arrival and coupling-gain trade-offs under a rotating star formation, (ii) aggressive compression and expansion stress-tests demonstrating the necessity of certified cooperative coupling to prevent transient collisions and network disconnections, and (iii) constrained narrow corridor traversal demonstrating arbitrary shape morphing and compatibility with existing safe planning schemes. In all scenarios, agents are initialized with random perturbations near their assigned stations satisfying the initial margin condition $\rho_0 < s(0)$. Global network connectivity for the star template is certified over a minimal 15-edge spanning backbone $\mathcal{H} = (\mathcal{V}, \mathcal{E}_{\mathcal{H}})$ comprised of the 5 inner pentagonal ring edges and 10 tip-to-valley spoke edges ($\lambda_2(L_{\mathcal{H}}) \approx 0.9316$, maximum unit backbone edge $\ell_{\mathcal{H},0} \approx 1.9497$), yielding effective disagreement decay rate $\gamma_{\mathcal{H}} = 1 + 0.1540m$. Target star stations rotate at angular rate ω around $r(t)$ and scale radially via $a(t) = a_\infty + (a_0 - a_\infty)e^{-\kappa t}$.

A. Finite-Time Tracking and Coupling-Gain Trade-offs

We first evaluate the influence of coupling gain $m \in \{0.0, 0.5, 2.0, 5.0, 10.0\}$ during rotating star compression ($\kappa = 2.0$). As shown in Figure 1 and Table II, all trajectories reach exact station tracking within the universal deadline $T \approx 1.3621$ s, empirically verifying that finite-time convergence is guaranteed independently of interaction topology and coupling strength.

TABLE II: Settling times and control metrics across coupling gains m under rotating star formation ($\kappa = 2.0$, $T = 1.3621$ s).

m	$t_{10^{-6}}$ (s)	T (s)	Peak $\|u\|_2$	J_u
0.0	1.1517	1.3621	16.9828	429.44
0.5	0.8117	1.3621	17.0860	429.92
2.0	0.5250	1.3621	18.0322	431.55
5.0	0.4467	1.3621	22.2186	434.95
10.0	0.3933	1.3621	32.8156	440.74

Cooperative coupling accelerates disagreement decay, reducing the settling time from 1.1517 s at $m = 0.0$ to 0.3933 s at $m = 10.0$. However, this introduces an engineering trade-off: higher gains require moderately increased peak control authority (rising from 16.98 to 32.82) while overall control effort J_u increases marginally (Table II).

B. Cooperative Coupling Necessity: Transient Safety and Connectivity

To demonstrate that cooperative coupling is indispensable for safe execution, the swarm is stress-tested under rapid star compression ($\kappa = 5.0$) and expansion ($\kappa = 8.0$), comparing certified coupling against uncoupled tracking ($m = 0.0$, $\gamma_{\mathcal{H}} = 1.0$) as detailed in Table III and Figure 2.

- **Compression ($\kappa = 5.0$, $\gamma_{\text{req}} = 4.4882$):** The star contracts from $a_0 = 1.70$ down to $a_\infty = 0.72$. For uncoupled agents ($m = 0.0$), tracking lag causes relative distortion to exceed nominal clearances, resulting in a physical collision breach ($d_{x,\min} < d_{\text{safe}}$) despite collision-free nominal stations. In contrast, certified coupling ($m = 30.0$) ensures $s(t) > \rho_{\mathcal{H}}(t)$ and guarantees collision avoidance ($d_{x,\min} > d_{\text{safe}}$).
- **Expansion ($\kappa = 8.0$, $\gamma_{\text{req}} = 4.1367$):** The star expands from $a_0 = 1.70$ up to $a_\infty = 3.02$, leaving a narrow terminal clearance margin of 0.112 m along the backbone. Uncoupled tracking suffers outward lag, breaching communication range ($\ell_{x,\max} > R_c$) for over 0.45 s. Certified coupling ($m = 25.0$) strictly preserves backbone links below R_c.

C. Arbitrary Formation Morphing and Constrained Corridor Traversal

Finally, we demonstrate the usability of the framework in practical constrained environments. The swarm traverses a narrow corridor ($x \in [12, 30]$ m, half-width $y_{\text{wall}} = 1.90$ m) by transitioning smoothly via C^2 quintic polynomial blending from an initial circular ring ($R = 3.60$ m) into a slender chevron ($y \in [-1.50, 1.50]$ m), recovering the ring upon exit (Figure 3). Spanning network connectivity is maintained along a designated 10-cycle backbone graph $\mathcal{H}$. As illustrated in Figure 3, agents achieve exact finite-time convergence ($T \approx 1.14$ s) and safely negotiate the passage without collision ($d_{x,\min} \geq 1.32$ m) or link loss ($\ell_{x,\max} \leq 2.22$ m), demonstrating seamless compatibility with collision-free formation planning algorithms.

VII. Conclusion

A first-order distributed controller has been presented for exact finite-time tracking of arbitrary prescribed station trajectories. With centered offsets, the swarm centroid reaches its reference and the requested relative formation is recovered by the same finite-time deadline. An additional disagreement estimate provides a tube around the prescribed pairwise distances. For a connected spanning backbone, this tube yields sufficient safety and communication-range conditions and a closed-form inequality for selecting the coupling gain. The simulations show that coupling can reduce transient formation distortion and that nominally safe target stations alone do not prevent transient collisions. They also expose the conservatism of the certificate for the reported ten-agent initialization, motivating experiments designed explicitly to satisfy the initial-margin condition.

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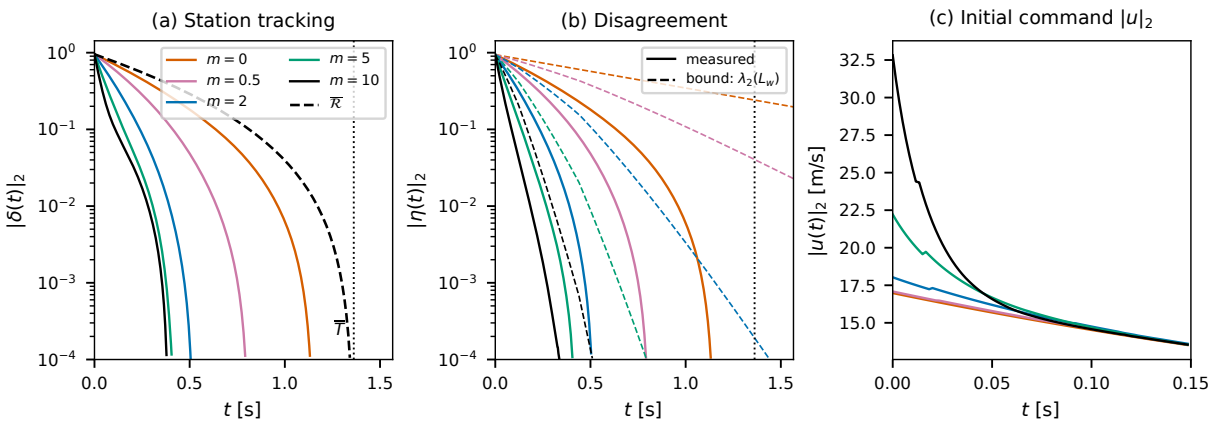
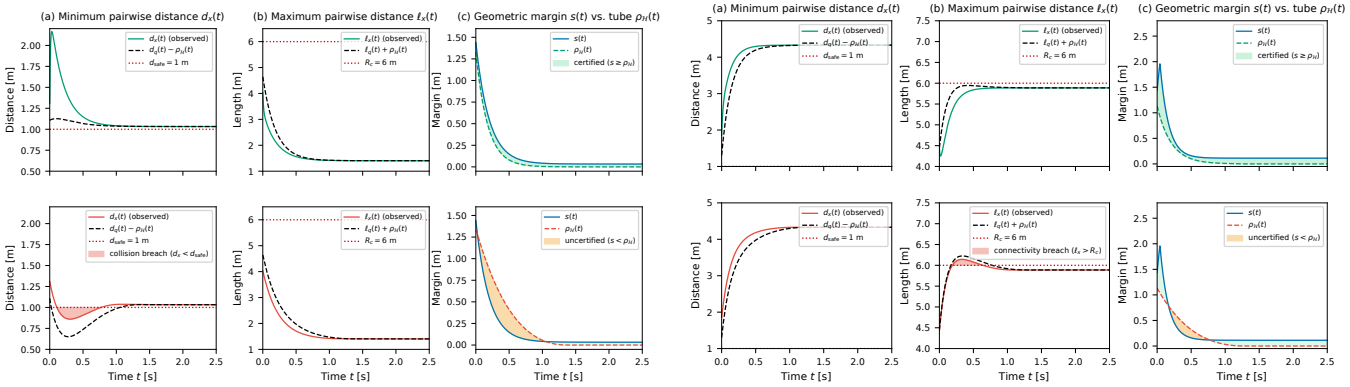


Fig. 1: Finite-time tracking, disagreement decay, and initial control effort across coupling gains m ($\kappa = 2.0$). (a) Station tracking $\|\delta(t)\|_2$ against theoretical bound $\overline{\mathcal{R}}(t)$ and deadline T. (b) Disagreement $\|\eta(t)\|_2$ against analytical envelope bounds. (c) Transient control command $\|u(t)\|_2$ over $t \in [0, 0.15]$ s.



(a) Formation compression ($\kappa = 5.0$).

(b) Formation Expansion ($\kappa = 8.0$).

Fig. 2: Transient safety and connectivity verification over $t \in [0, 2.5]$ s. In both panels, columns show the minimum pairwise separation $d_x(t)$ (left), maximum pairwise distance $\ell_x(t)$ (middle), and geometric margin $s(t)$ versus disagreement tube $\rho_H(t)$ (right). **Left:** Star compression ($\kappa = 5.0$); top row: Certified Safe ($m = 30.0$), bottom row: Unsafe Mid-Transient Collision ($m = 0.0$). **Right:** Star expansion ($\kappa = 8.0$); top row: Certified Safe & Connected ($m = 25.0$), bottom row: Unsafe Transient Disconnection ($m = 0.0$).

TABLE III: Comparison of certified safe/connected vs. uncoupled baseline under aggressive star maneuvers ($t \in [0, 2.5]$ s).

Maneuver	Case	m	κ	$\gamma_{\mathcal{H}}$	γ_{req}	t_{set} (s)	$d_{x,\min}$	$\ell_{x,\max}$	Peak $\|u\|_2$	J_u	Outcome
Compression	Safe	30.0	5.0	5.6194	4.4882	0.3800	1.0326	4.0338	88.2953	513.97	Certified; Safe, Connected
	Unsafe	0.0	5.0	1.0000	4.4882	1.1517	0.8605	4.0338	35.0544	480.66	Uncertified; Safety Violation
Expansion	Safe	25.0	8.0	4.8495	4.1367	0.4183	1.8550	5.8880	83.5699	3624.11	Certified; Safe, Connected
	Unsafe	0.0	8.0	1.0000	4.1367	1.1333	1.8550	6.1424	72.3516	3620.46	Uncertified; Connectivity Violation

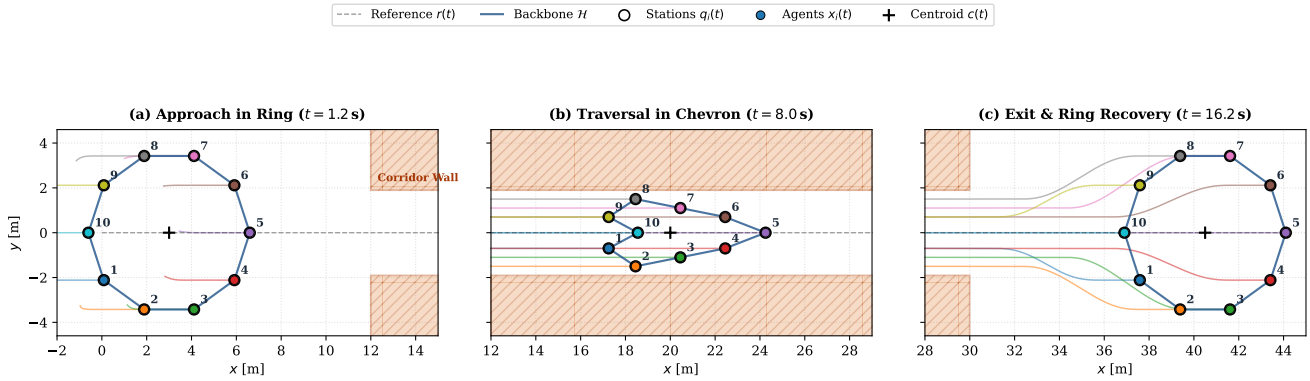


Fig. 3: Arbitrary formation morphing and constrained corridor traversal ($n = 10$, $m = 10.0$). Snapshots depict: (a) approach and finite-time convergence in ring geometry ($t = 1.2$ s), (b) morphing into a slender chevron to navigate the narrow corridor ($t = 8.0$ s), and (c) exit and ring recovery ($t = 16.2$ s). Target stations q_i (circles), agent positions x_i (colored dots), continuous trajectories, reference centroid $r(t)$ (dashed line), and the 10-cycle backbone $\mathcal{H}$ (solid blue edges) remain strictly collision-free ($d_{x,\min} \geq 1.32$ m) and connected ($\ell_{x,\max} \leq 2.22$ m).